KV Cache Steering for Controlling Frozen LLMs

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Abstract

We propose *cache steering*, a lightweight method for implicit steering of language models via a one-shot intervention applied directly to the key-value cache. To validate its effectiveness, we apply cache steering to induce chain-of-thought reasoning in small language models. Our approach constructs steering vectors from reasoning traces, obtained either from teacher models (e.g., GPT-4o) or existing human annotations, that shift model behavior toward more explicit, multi-step reasoning without fine-tuning or prompt modifications. Experimental evaluations on diverse reasoning benchmarks demonstrate that cache steering improves both the qualitative structure of model reasoning and quantitative task performance. Additional experiments show that the method also scales to larger models and yields further gains on challenging datasets such as GPQA and MATH. Compared to prior activation steering techniques that require continuous interventions, our one-shot cache steering offers substantial advantages in terms of inference latency, hyperparameter stability, and ease of integration with existing inference APIs. Beyond mere reasoning induction, we show that cache steering enables controllable transfer of reasoning styles (e.g., stepwise, causal, analogical), making it a practical tool for behavior-level guidance of language models.

1 Introduction

The ability of large language models to perform complex reasoning is a key driver of their increasing utility. However, this potential is not always spontaneously realized, especially in smaller models which may possess latent reasoning capabilities that require specific guidance to activate. Traditional methods for uncovering these abilities, such as supervised fine-tuning or few-shot prompting with chain-of-thought examples, can be effective but often demand significant data or intricate prompt design. The question then arises: can we develop more lightweight interventions to unlock and steer these inherent reasoning processes post-training?

One promising direction is activation steering (Turner et al., 2024; Rimsky et al., 2024), which aims to guide model behavior by directly modifying its internal hidden states. While promising for its ability to influence outputs without retraining, activation steering often requires continuous interventions at each token generation step throughout the decoding process to be effective (Wehner et al., 2025). This continuous manipulation can introduce instability, making the outcomes highly sensitive to hyperparameter choices (e.g., targeted layers, intervention strength) and potentially leading to a degradation in generation quality.

To address these issues, we introduce a method called **cache steering**. Our approach operates by making a targeted, one-time modification directly to the key-value cache of a Transformer model, typically after

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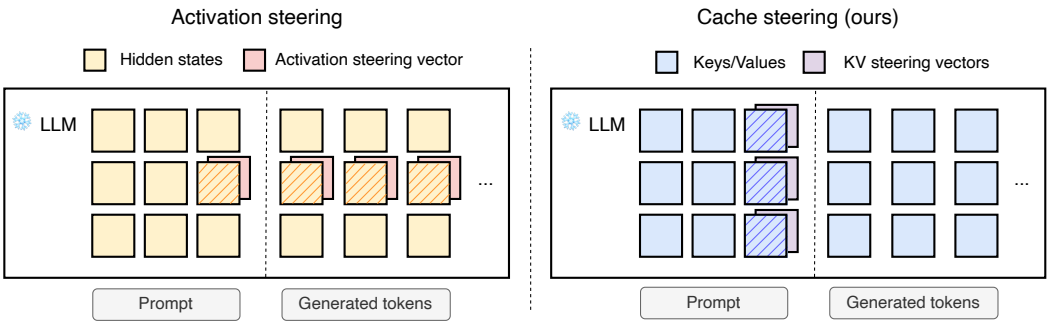

Figure 1: **Activation steering vs. Cache steering**. Activation steering (left) injects vectors into hidden states *dynamically* during decoding, typically at a single chosen layer. Intervening at multiple layers is possible but often amplifies effects across the network, making the method sensitive to tuning and prone to instability. Cache steering (right) instead modifies the pre-computed KV cache once after the prefilling step. Because these cached representations are *fixed*, the intervention can be applied consistently across all layers and then implicitly influences future tokens, leading to stable and efficient inference without repeated runtime injections.

the cache has been populated by an initial prompt. By applying steering vectors, derived either from reasoning traces generated by a capable teacher model like GPT-4o or from existing human/dataset annotations, to these cached key and value representations, we can guide the reasoning trajectory of language models. This single intervention, applied before token generation begins, effectively steers the model towards more explicit, multi-step reasoning without altering model weights or requiring complex prompt modifications. Compared to activation steering, which applies interventions at every decoding step, cache steering avoids cascading effects, is robust to hyperparameter choices, introduces virtually no runtime cost, and seamlessly integrates with standard inference pipelines.

We demonstrate that our method improves reasoning structure and, in many cases, task accuracy on multiple benchmarks, including GSM8K, ARC-Challenge, CSQA, and PIQA, and further scales to larger models and harder datasets such as GPQA and MATH. Beyond simple reasoning induction, cache steering enables *controllable transfer of reasoning styles* (e.g., stepwise, causal, analogical), illustrating its value as a practical tool for behavior-level guidance.

Overall, our key contributions are as follows:

- We propose **cache steering**, a one-shot modification of the KV cache that provides a lightweight and production-ready alternative to activation steering and fine-tuning.
- We demonstrate that cache steering can **distill reasoning styles** from teacher models or existing reasoning traces into smaller models without weight updates or prompt augmentation.
- We conduct extensive evaluations across multiple model families and benchmarks, analyze efficiency and stability, and provide additional results on larger models and challenging reasoning datasets.

2 Related Work

Reasoning and Chain-of-Thought prompting. A widely adopted approach to enhance reasoning abilities in LLMs involves demonstrating example solutions to the problem (in-context learning or ICL) that contain a step-by-step reasoning process (Chain-of-Thought or CoT) in a prompt to the language model (Brown et al., 2020; Wei et al., 2022), a technique known as few-shot prompting. Zero-shot variants of CoT prompting simplify this approach by adding instructions such as "Let's think step by step" to elicit step-by-step reasoning without the need for example demonstrations (Kojima et al., 2022).

The code is available at `https://github.com/MaxBelitsky/cache-steering`.

Recent work shows that reinforcement learning can lead to remarkable reasoning capabilities, which can be effectively distilled into smaller models through supervised fine-tuning (Guo et al., 2025). These findings suggest that it is not enough to just trigger CoT reasoning in the language models, but the **style** of the reasoning matters. This motivates our approach, which aims to directly steer small models toward reasoning behavior reminiscent of larger teacher models via cache-level interventions.

Activation steering. Activation steering, also known as representation engineering, is a technique used to control the generation process of LLMs implicitly by manipulating their intermediate activations during decoding, typically through linear interventions (Rimsky et al., 2024; Turner et al., 2024). Multiple works have applied activation steering to induce or suppress specific behaviors in models without retraining. The examples include sentiment, topic and style control (Turner et al., 2024); function steering (Todd et al., 2024; Postmus & Abreu, 2024), removing or inducing refusal behavior, (Lee et al., 2025), toxicity reduction (Turner et al., 2024), truthfulness (Wang et al., 2025), editing factual knowledge (Yin et al., 2024), reasoning (Zhang & Viteri, 2025; Galichin et al., 2025) and other (Wehner et al., 2025).

In its most basic form, activation steering involves two steps: vector extraction and injection of the vector into the activations of the model at inference. The vector extraction stage involves computing a "steering vector", which is commonly done by aggregating activations from pairs of positive prompts with desired behavior and negative or sometimes neutral prompts, forming a contrastive set $C=\{(p_0^+,p_0^-),(p_1^+,p_1^-),\ldots,(p_N^+,p_N^-)\}$. The most common aggregation method is Difference-in-Means (Wehner et al., 2025), which is identical to Mean-of-Differences when the vectors are paired:

$$\boldsymbol{s}_l = \frac{1}{N} \sum_{(p^+,p^-)\in C} f_l(p^+) - f_l(p^-)$$

where f_l represents the part of the Transformer model (e.g., the whole decoder layer) at layer l and N is the number of examples in the contrastive dataset C. To steer the model's output, the steering vector is added to the activations of specific layers during inference:

$$\boldsymbol{h}_l^* = \boldsymbol{h}_l + c\boldsymbol{s}_l$$

where $\boldsymbol{h}_l$ represents the activations at layer l before steering, $\boldsymbol{s}_l$ is a steering vector extracted from layer l, and c is a coefficient that determines the strength of the steering.

It is important to mention that the vector can be extracted and applied to different token positions, layers, and parts of the model, which are treated as hyperparameters or design choices. Usually, it is a common practice to perform a grid search to determine the layers to apply steering to and the value of the steering strength coefficient c (Turner et al., 2024; Lee et al., 2025; Wang et al., 2025; Dong et al., 2024; Rimsky et al., 2024; Wang et al., 2024; Stolfo et al., 2025; Zhang & Viteri, 2025; Postmus & Abreu, 2024).

While activation steering offers a tool for model control, it typically requires continuous intervention during generation (Wehner et al., 2025), which can be expensive and can lead to unstable generations. In contrast, our work extends the activation steering paradigm to the key-value (KV) cache, enabling a one-shot intervention that is both more efficient and more stable at inference time.

Cache manipulation. Another emerging line of research explores the idea of modifying the key-value (KV) cache from the memory and efficiency perspective (Li et al., 2024; Liu et al., 2025a; Ge et al., 2024; Mu et al., 2023). These approaches aim to reduce the memory footprint or compress contextual representations through KV cache manipulation. Building on this idea, Liu et al. (2025b) introduced a method for augmenting the KV cache to improve the performance on tasks that require reasoning abilities. The authors use a differentiable "coprocessor", which allows augmenting the KV cache as a pre-generation step instead of modifying activations directly during the forward pass. However, in order to augment the KV cache, the method requires training a separate model, which makes this method less practical than pure activation steering methods introduced in the previous subsection. In contrast, our approach aims to use the KV cache as a target for behavioral control in small models without training auxiliary modules.

3 Cache Steering

We introduce **cache steering**, a lightweight method for inducing structured reasoning in language models by applying steering vectors directly to the key-value cache. Unlike traditional activation